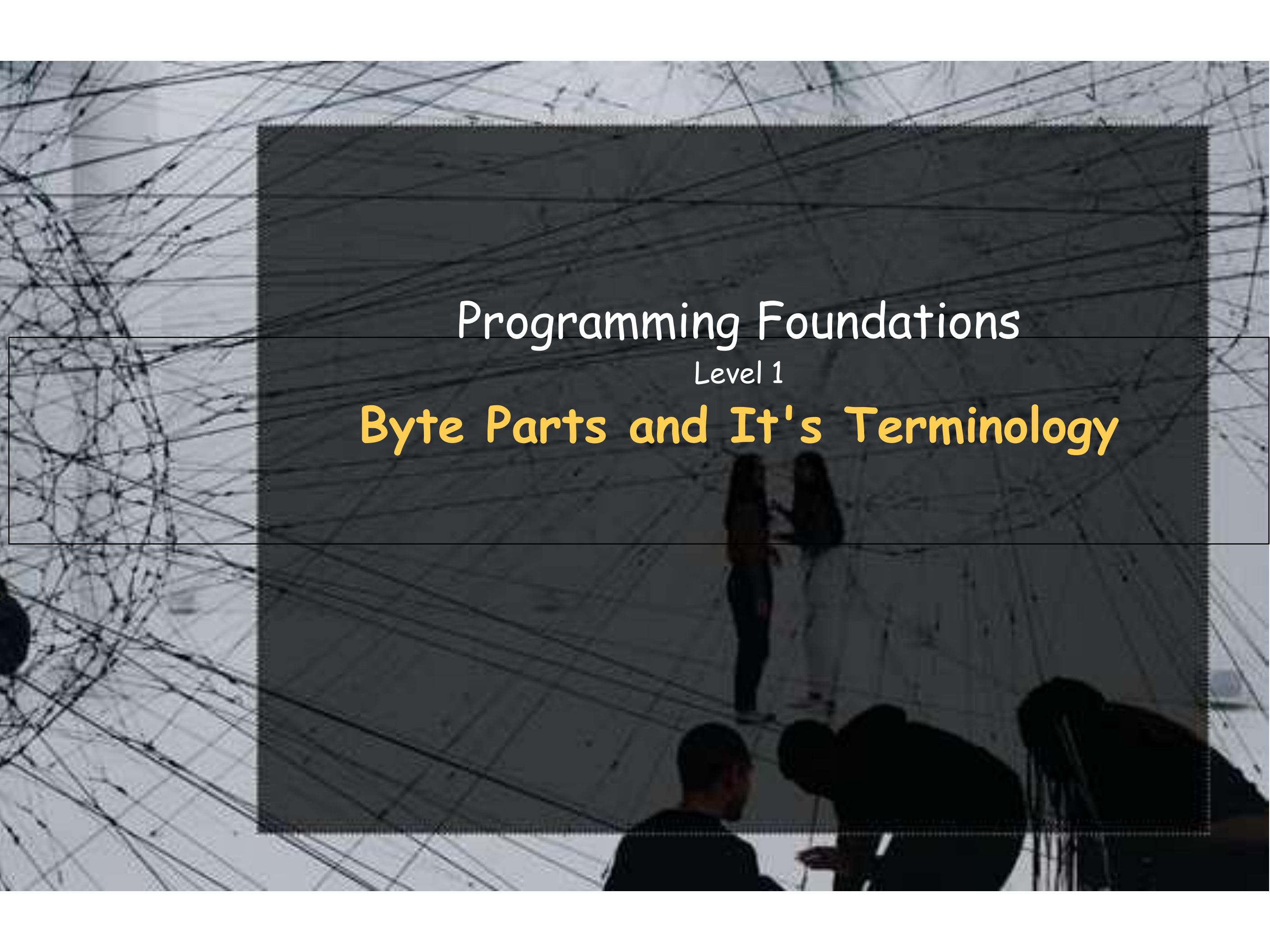




Programming Foundations Level 1

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Programming Foundations

Level 1

Byte Parts and It's Terminology

Byte Parts: Bit, Crumb, Nibble, Byte — LSB & MSB

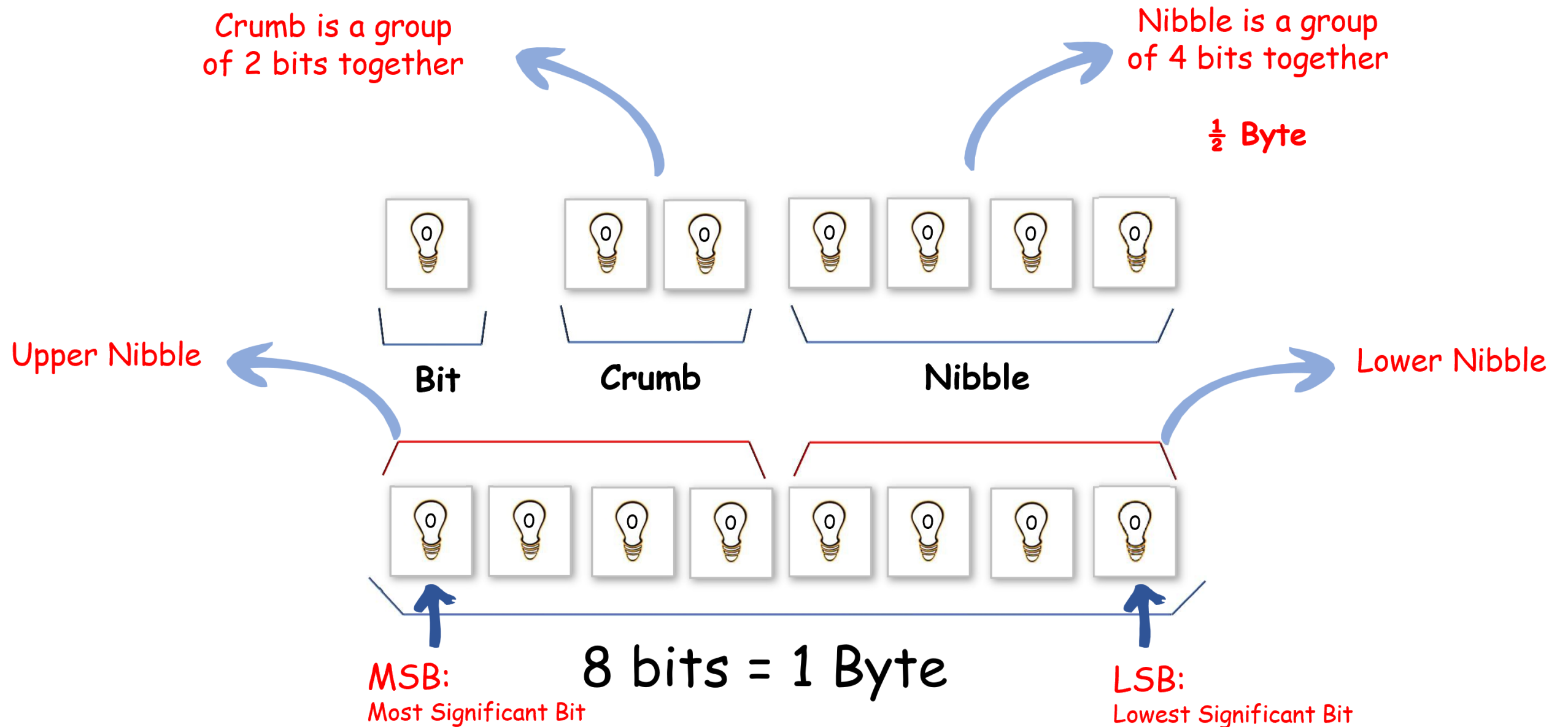
Every piece of digital data—text, images, sound—is built from **bits** (0 or 1).

Bits group into larger, human-friendly units with specific names: **crumbs** (2 bits), **nibbles** (4 bits), and **bytes** (8 bits).

Understanding these building blocks—and how **LSB** and **MSB** work—makes binary, hex, and memory maps much easier.

Core Definitions

Byte Parts:



Byte Parts:

- **Bit:** the smallest unit of information — **0 or 1**.
- **Crumb:** a group of **2 bits**. (*Yes, “crumb” is the official term used in your slides.*)
- **Nibble:** a group of **4 bits** ($\frac{1}{2}$ byte).
- **Byte:** **8 bits**. (1 byte = 2 nibbles = 4 crumbs.)

Lower vs Upper Nibble

- **Lower Nibble:** the rightmost 4 bits of a byte (b3 b2 b1 b0).
- **Upper Nibble:** the leftmost 4 bits of a byte (b7 b6 b5 b4). These names matter because hex maps perfectly to nibbles: 1 byte $\leftrightarrow$ 2 hex digits (upper nibble = first hex digit, lower nibble = second).

LSB & MSB

- **LSB (Least Significant Bit):** the rightmost bit of the byte (b0). It carries the smallest weight (2^0).
- **MSB (Most Significant Bit):** the leftmost bit of the byte (b7). It carries the largest weight (2^7). We call them "least/most significant" because they contribute the least/most to the numeric value of the binary number.

Quick Examples

■ Byte: 0101 1100

■ Upper nibble = 0101 (hex 5)

■ Lower nibble = 1100 (hex C)

■ LSB = 0 (rightmost)

■ MSB = 0 (leftmost)

■ Byte: 1010 0111

■ Upper nibble = 1010 (hex A)

■ Lower nibble = 0111 (hex 7)

■ LSB = 1, MSB = 1

Interconnection

- ◆ 2 bits = **Crumb**, 4 bits = **Nibble**, 8 bits = **Byte** → hierarchical building blocks.
- ◆ **Upper nibble** (left 4) + **Lower nibble** (right 4) = 1 **byte** → aligns perfectly with **two hex digits**.
- ◆ **MSB** (leftmost) carries the **highest weight**, **LSB** (rightmost) the **lowest** → drives numeric value.

By mastering crumbs, nibbles, bytes, and LSB/MSB, you read binary/hex confidently and avoid off-by-nibble errors in real code! 🚀

Activity

Instructions:

Answer the following questions based on what you learned about **bits**, **crumbs**, **nibbles**, **bytes**, and their parts (**LSB/MSB**):

- 1 What is a **bit**?
 - 2 What is a **crumb**?
 - 3 What is a **nibble**?
 - 4 What is a **byte**?
 - 5 How many **bits** are in a byte?
 - 6 How many **crumbs** are in a byte?
 - 7 How many **nibbles** are in a byte?
 - 8 How many **crumbs** are in a nibble?
 - 9 What is the **lower nibble**?
 - 10 What is the **upper nibble**?
- 1 1 What is **LSB**, why do we call it **LSB**, and what is its value?
 - 1 2 What is **MSB**, why do we call it **MSB**, and what is its value?

Activity Solution

Solution:

1 What is a **bit**?

Bit: The smallest unit of information in computing; it can be 0 or 1.

2 What is a **crumb**?

Crumb: A small group of 2 bits.

3 What is a **nibble**?

Nibble: A group of 4 bits (half of a byte).

4 What is a **byte**?

Byte: A collection of 8 bits (2 nibbles or 4 crumbs).

5 How many **bits** are in a byte?

Bits in a byte: 8 bits.

6 How many **crumbs** are in a byte?

Crumbs in a byte: 4 crumbs (8 bits ÷ 2 bits per crumb).

7 How many **nibbles** are in a byte?

Nibbles in a byte: 2 nibbles (8 bits ÷ 4 bits per nibble).

8 How many **crumbs** are in a nibble?

Crumbs in a nibble: 2 crumbs (4 bits ÷ 2 bits per crumb).

9 What is the **lower nibble**?

Lower nibble: The rightmost 4 bits of a byte (least significant half).

10 What is the **upper nibble**?

Upper nibble: The leftmost 4 bits of a byte (most significant half).

1 1 What is **LSB**, why do we call it LSB, and what is its value?

LSB (Least Significant Bit):

It's the **rightmost** bit in a binary number.

Called "least significant" because it has the **smallest weight** ($2^0 = 1$).

Its value contributes the least to the overall binary value.

1 2 What is **MSB**, why do we call it MSB, and what is its value?

MSB (Most Significant Bit):

It's the **leftmost** bit in a binary number.

Called "most significant" because it has the **highest weight** ($2^7 = 128$ in an 8-bit byte).

Its value contributes the most to the binary value, often used to indicate **sign** or **priority**.

🏁 Conclusion

- ✅ **Bit** → **Crumb (2)** → **Nibble (4)** → **Byte (8)** are the fundamental groupings of binary data.
- ✅ **Upper/Lower nibble** make byte parsing and hex representation straightforward.
- ✅ **LSB/MSB** determine each bit's weight and are key in numeric value and flag interpretation.

Mastering these byte parts turns "scary binary" into a clean, predictable structure you can read and write with confidence. 💪

Thank you 😊